

ROHAN PAREEK

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EDUCATION

University of Illinois at Urbana-Champaign (UIUC), Champaign, IL

Master of Computer Science at Grainger College of Engineering

Bachelor in Economics, Minor in Computer Science

Dec 2027

May 2026

Clubs: Association of Computing Machinery, Illini Electric Motorsport, FACES Consulting, Econ Data Lab, UIUC Economics Club

Relevant Coursework: Data Structures & Algorithms, Systems Programming, Database Systems, Cloud Computing Applications, Applied Machine Learning, Text Information Systems, Probability & Statistics for Computer Science, Data Visualization, Linear Algebra

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C++, TypeScript, JavaScript, MATLAB, C, HTML, CSS

Frameworks & Libraries: Pandas, NumPy, statsmodels, scikit-learn, PyTorch, Jupyter, regression, hypothesis testing, LangChain

Tools & Platforms: PostgreSQL, SQLite, Git, Docker, AWS, Linux/Unix, FastAPI, Chart.js, Matplotlib, REST APIs, OpenAI API, Redis

Domains: Statistical Modeling, Time-Series Forecasting, Data Pipelines & ETL, Exploratory Data Analysis, Data Visualization, AI & ML

EXPERIENCE

Siebel School of Computing and Data Science

ML Research Assistant, Prof. Pablo D. Robles Granda

Champaign, IL

Sept 2025 – Present

- Developing an end-to-end fMRI pipeline (nilearn, PyTorch Geometric) converting raw brain scans into functional-connectivity graphs and training Graph Convolutional Networks to classify Alzheimer's, comparing Pearson, Spearman, and Kendall connectivity methods
- Built a multimodal behavioral analysis pipeline for the StudentLife dataset, engineering sliding-window sequences from continuous sensor streams and training LSTM temporal models to predict well-being and academic outcomes

Amazon Web Services (AWS)

Software Development Engineering Intern

Portland, OR

May 2025 – Aug 2025

- Developed an internal developer tool that renders AWS CloudFormation templates as navigable, interactive dependency graphs, in TypeScript and React within a shared monorepo architecture
- Implemented graph editing with node deletion, pending-state previews, and confirmation workflows, enabling users to model infrastructure changes before applying them
- Architected event-driven canvas state management using a type-safe event bus, decoupling rendering from user interaction logic

Swing Phi LLC

Software Engineering Intern

Nashville, TN

Dec 2024 – Apr 2025

- Built end-to-end financial data pipeline in Python ingesting and processing 50+ unstructured earnings reports, news articles, and SEC filings via Yahoo Finance API, applying NLP techniques to generate quantitative trading signals
- Designed and trained image classification models in PyTorch to detect technical chart indicators (MACD, RSI, VWAP) from 5 years of historical price data, delivering structured feature outputs to downstream trading strategy models

RELEVANT PROJECTS

CoinCast | Python, FastAPI, statsmodels, Pandas, NumPy, React, Chart.js, CoinGecko API

- Built a time-series forecasting service for 10+ cryptocurrencies, fitting Holt's exponential smoothing models to multi-year daily price histories and serving point forecasts with confidence intervals over configurable 7-30 day horizons via a FastAPI REST endpoint
- Developed an interactive React dashboard rendering historical series against forecast bands in Chart.js, enabling side-by-side comparison of predicted trajectories and realized prices across assets and horizons

PrivateMind | Python, FastAPI, SQLAlchemy, ChromaDB, sentence-transformers, Ollama, React, TypeScript, Docker Compose

- Architected a fully offline AI assistant running local LLM inference via Ollama and in-process sentence-transformer embeddings, enabling document ingestion, semantic search, and SM-2 spaced repetition with no data leaving the machine
- Developed a containerized full-stack system with FastAPI backend and React/TypeScript frontend deployed via Docker Compose, implementing an LLM-as-a-Judge eval harness with lint and unit test gates to validate retrieval quality before each milestone shipped

FridgeVision | Python, Flask, OpenCV, YOLO, OpenAI API, HTML, JavaScript

- Built an accessibility-focused web application enabling visually impaired users to photograph a fridge and query item locations in natural language, combining OpenCV shelf detection and YOLO object detection with an LLM for query understanding, achieving over 90% detection accuracy
- Implemented Flask backend orchestrating the image-to-inventory pipeline, chaining upload, shelf segmentation, and item detection into persistent fridge state that powers inventory tracking, shopping lists, and recipe generation

LEADERSHIP

Technical Project Manager & Director @ FACES Consulting

Aug 2025 – May 2026

- Led consultants to conduct AI research for a cybersecurity firm developing a next-generation TDIR platform with CTEM capabilities
- As a Project Director, trained and led 6+ Senior Consultants and Project Managers to effectively work with a large team of consultants and companies ranging from Education to AI and Cybersecurity